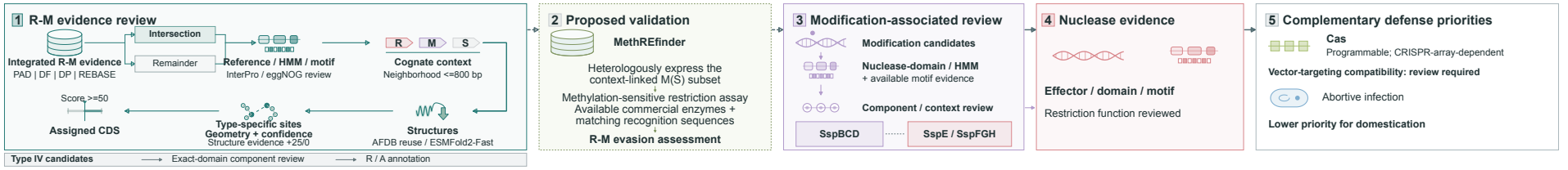
a Evidence criteria for DNMB-guided barrier prioritization



b Comparative defense landscape across thermophilic genomes

① R-M evidence and curation · ② Methylation-associated evidence · ③ Non-canonical DNA modification system · ④ Nuclease-associated evidence · ⑤ Complementary defense priorities

R-M CDSs in PAD DF DP · Hits REB · Intersection (n/R-M pool) · R-M subunits and contexts: I CDS R M S MS RMS; I Groups; II CDS R M; III CDS R M M M MR; IV R A · Assigned I–III CDSs (R/total) · Non-canonical Annotated CDSs PAD DF DP · Exclusive CDSs R-M-excluded union (n/pool) · Modification · Nuclease · Cas · Abortive infection · Other · Unresolved · Strains

R-M CDSs in PAD	DF	DP	Hits REB	Intersection (n/R-M pool)	Assigned I–III CDSs (R/total)	Non-canonical Annotated CDSs PAD	DF	DP	Exclusive CDSs R-M-excluded union (n/pool)	Strains
8	9	8	20	4/21	3/ 8	35	24	23	19/40	*G. caldolyticus* NEB414
10	13	16	23	10/23	3/11	27	13	14	6/34	*Geobacillus* genomosp. 3
8	8	13	23	6/23	3/ 8	25	13	15	5/36	***G. kaustophilus* HTA426**
5	8	13	23	5/23	2/ 7	34	18	27	11/50	*G. kaustophilus* NBRC 102445
8	10	12	20	8/20	3/ 9	43	30	30	22/52	*G. kaustophilus* T-1
6	9	14	21	6/21	2/ 9	56	39	38	22/73	*G. proteiniphilus* T-6
8	10	13	20	7/20	3/ 9	35	16	18	10/44	*Geobacillus* sp. C56-T3
10	12	12	24	9/24	3/10	38	30	33	22/52	*Geobacillus* sp. Y412MC52
6	7	11	17	6/17	2/ 7	38	17	26	12/52	*Geobacillus* sp. GHH01
	3	5	13	0/13	0/ 4	48	27	43	17/72	*Geobacillus* sp. JS12
7	10	13	20	7/20	2/ 8	28	22	26	17/36	*Geobacillus* sp. 46C-IIa
2	3	3	8	2/ 8	0/ 1	16	5	12	2/25	*Geobacillus* sp. 47C-IIb
5	5	9	19	5/19	2/ 6	37	21	31	13/57	*G. stearothermophilus* DSM 458
6	6	11	18	6/18	2/ 7	44	26	28	17/55	*G. stearothermophilus* B5
9	11	14	21	9/21	3/ 9	46	30	37	18/67	*G. stearothermophilus* DG-1
2	6	6	13	2/13	2/ 5	19	11	13	6/28	***G. stearothermophilus* ATCC 12980**
9	9	11	20	7/20	3/ 8	45	21	26	13/53	***G. stearothermophilus* SJEF4-2**
9	12	16	23	9/23	4/11	46	26	37	18/65	***G. stearothermophilus* EF60045**
9	9	8	21	7/22	2/ 7	35	17	16	9/39	*G. stearothermophilus* EF60063
8	9	13	23	7/23	3/10	30	12	30	9/48	*G. stearothermophilus* EF60133
8	15	19	28	8/28	4/14	44	23	45	18/66	*G. stearothermophilus* EF60134
13	14	15	25	12/25	4/13	37	24	30	17/51	*G. subterraneus* KCTC3922
11	13	17	29	11/29	4/14	45	37	48	31/64	*G. subterraneus* E55-1
7	7	10	20	5/20	2/ 7	25	15	19	8/38	*G. subterraneus* CPW16
8	10	13	21	8/21	3/ 9	23	12	21	10/33	*G. subterraneus* OXR-76
	1	8	17	0/17	0/ 2	73	56	59	44/88	*G. thermocatenulatus* KCTC 3921
6	7	10	16	6/16	3/ 7	8	5	15	0/23	*G. thermodenitrificans* NG80-2
	2	2	8	0/ 8	0/ 1	8	4	8	2/12	***G. thermodenitrificans* KCTC3902**
4	5	5	14	3/14	2/ 5	17	7	16	4/24	*G. thermodenitrificans* T12
7	8	7	15	6/15	2/ 6	19	10	14	2/31	*G. thermodenitrificans* ID-1
2	4	6	13	2/13	0/ 3	25	11	18	5/33	*G. thermodenitrificans* K1041
7	9	13	24	6/24	3/ 9	28	16	18	5/42	*G. thermoleovorans* CCB_US3_UF5
3	5	11	21	3/21	2/ 5	30	14	25	8/48	*G. thermoleovorans* FJAT-2391
7	9	13	24	6/24	3/ 9	28	16	18	5/42	*G. thermoleovorans* ARTRW1
5	8	17	30	5/30	2/ 8	20	10	23	5/41	*G. thermoleovorans* SURF-48B
3	4	7	13	3/13	2/ 5	39	19	20	9/51	*G. thermoleovorans* V0
4	5	6	17	3/17	1/ 4	21	8	16	4/35	***G. thermoleovorans* KCTC 3570**
3	4	6	17	2/17	3/ 6	45	25	28	18/55	*G. zalihae* SURF-189
6	11	16	29	6/29	4/10	53	27	60	19/89	*G. zalihae* SURF-114
7	8	9	21	5/21	3/ 7	46	21	25	14/58	*G. zalihae* JS1-2
	3	3	14	0/14	0/ 1	12	10	20	6/27	*P. caldoxylosilyticus* ER4B
2	3	2	13	2/13	1/ 2	14	9	22	7/28	*P. caldoxylosilyticus* KH1-5
2	3	2	13	2/13	1/ 2	15	10	22	8/28	*P. caldoxylosilyticus* KH1-6
	1	1	11	0/11	0/ 0	20	16	27	12/34	*P. caldoxylosilyticus* KH3-5
3	2	2	11	1/12	1/ 3	22	17	26	14/34	*Parageobacillus* sp. KH3-4
12	12	16	25	12/25	4/14	48	36	46	27/64	*P. thermoglucosidasius* C56-YS93
2	4	9	15	2/15	1/ 4	32	26	40	19/51	*P. thermoglucosidasius* DSM 2542
2	4	9	15	2/15	1/ 4	33	26	38	19/50	*P. thermoglucosidasius* NCIMB 11955
2	4	9	15	2/15	1/ 4	33	26	40	19/52	*P. thermoglucosidasius* TM242
4	6	10	18	4/18	2/ 8	35	31	40	24/52	*P. thermoglucosidasius* BH4-1
1	2	5	12	0/12	0/ 0	20	13	18	7/32	*P. toebii* NBRC 107807
3	3	5	14	0/14	1/ 2	25	15	18	6/35	*P. toebii* NEB718

Modification: BREX, DISARM, DMS_other, DndABCDE, PbeABCD (PT), SspBCD (PT), SspE / SspFGH (PT)

Nuclease: Gabija, Wadjet II, Wadjet III, Wadjet other, Zorya, Lamassu, Nhi, DARNA, Hma, MazEF, RloC, Schlafen Ig-like, CoCoNut, Mokosh II, pAgo Long-A, qatABCD, HEC (GmrSD-like)

Abortive infection: CRISPR-Cas, CBASS, AbiD, AbiE, Aristaios, DarTG, Pycsar, Ruguts, SoFic, Stk2, pAgo Long-B / Sir2, PD

Other: Kiwa, Ogmios, dXTPase

Unresolved: Ceres, Druantia, Menshen, gcu233, PDC-M, PDC-S, DP-only unassigned, Former R-M annotations

PAD only (647 CDSs) · DF only (114 CDSs) · DP only · PAD + DF · PAD + DP · DF + DP · n/pool

Coding sequences (CDSs): 1 · 2-3 · 4-7 · 8-15 · 16+

Subunit groups: (n) One symbol + group count · All groups R-supported · Any REBASE-linked group · M(S)-only groups

Family evidence: Putative nuclease-associated CDSs · Effector CDSs

R: nuclease · A: auxiliary

Genus: *G. = Geobacillus* · *P. = Parageobacillus*

Species: *G. caldoxylosilyticus* · *G. genomosp.* · *G. kaustophilus* · *G. proteiniphilus* · *G. stearothermophilus* · *G. subterraneus* · *G. thermocatenulatus* · *G. thermodenitrificans* · *G. thermoleovorans* · *G. zalihae* · sp. (unassigned) · *P. caldoxylosilyticus* · *P. thermoglucosidasius* · *P. toebii*

Annotations: Other · Mechanism not assigned yet · No Wadjet II

c CDS inclusion by assigned R-M type

R(n): incomplete context · M/S(n): included orphans · (—): not assessed · —: unavailable or not applicable · □ separately analyzed external case; not in cohort totals

***G. stearothermophilus* EF60045**

Type	Pool	Tool calls in pool (any family) DF	PAD	DP	DNMB	DNMB components R	M	S	A
I	9	9	9	9	9	3	3	3	—
II	0	0	0	0	0	0	0	—	—
III	3	2	2	2	2	1 (1)	1 (1)	—	—
IV	3	1	1	3	3	2	—	—	1
Total	**15**	**12**	**12**	**14**	**14**	**6 (1)**	**4 (1)**	**3**	**1**

***G. stearothermophilus* SJEF4-2**

Type	Pool	Tool calls in pool (any family) DF	PAD	DP	DNMB	DNMB components R	M	S	A
I	7	7	7	6	5	2 (1)	2	1	—
II	2	2	2	1	2	1	1	—	—
III	1	0	0	0	1	0	1 (1)	—	—
IV	2	0	1	1	2	2	—	—	0
Total	**12**	**9**	**10**	**8**	**10**	**5 (1)**	**4 (1)**	**1**	**0**

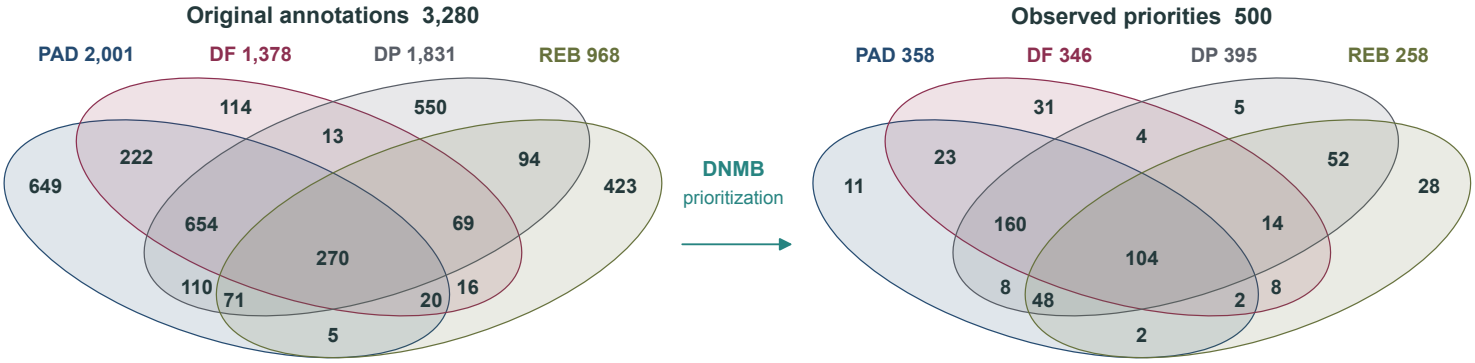
***B. coagulans* DSM 1**

Type	Pool	Tool calls in pool (any family) DF	PAD	DP	DNMB	DNMB components R	M	S	A
I	6	6	6	6	6	2	2	2	—
II	1	0	0	0	1	0	1 (1)	—	—
III	2	2	2	2	2	1	1	—	—
IV	3	1	1	3	3	2	—	—	1
Total	**12**	**9**	**9**	**11**	**12**	**5**	**4 (1)**	**2**	**1**

DNMB integrated defense comparison

52 genomes · Coding sequences (CDSs)

a DNMB prioritization



Areas schematic

R subunits + nuclease effectors

b Defense categories

Category	Number of genomes	DNMB	Putative nuclease-associated CDSs		DF	PAD	DP
			Total	Effector CDSs			
R-M	52	540	422	232	365	333	425
Modification	31	127	63	10	61	127	82
Nuclease	51	352	329	132	287	283	227
CRISPR-Cas	42	501	481	104	467	500	394
Abortive infection	50	213	23	6	118	204	36
Other mechanisms	7	15	0	0	13	15	0
Unresolved	52	1,192	15	15	73	558	662
Former R-M residual	11	12	1	1	0	4	11

c Defense families

Family / pool	Number of genomes	DNMB	Putative nuclease-associated CDSs		DF	PAD	DP
			Total	Effector CDSs			
R-M (current union)	52	540	422	232	365	333	425
BREX	8	48	48	8	37	48	45
DISARM	1	5	5	1	5	5	5
DMS_other	18	26	3	1	2	26	11
DndABCDE	7	37	0	0	7	37	14
PbeABCD (PT)	1	1	1	0	0	1	1
SspBCD (PT)	4	12	6	0	12	12	8
SspE / SspFGH (PT)	4	10	6	6	10	10	10
Gabija	9	20	20	10	20	20	17
Wadjet II	33	132	132	33	131	132	130
Wadjet III	4	16	16	4	16	16	12
Wadjet other	4	6	4	2	0	6	2
Zorya	11	45	45	11	18	45	4
Lamassu	3	9	9	3	9	9	9
Nhi	6	6	0	0	0	6	6
DARNA	1	1	1	1	1	1	1
Hma	1	3	3	1	3	3	3
MazEF	29	60	60	30	60	0	0
RloC	2	2	2	2	2	2	2
Schlafen Ig-like	1	1	1	1	1	0	0
CoCoNut	7	8	4	4	8	0	7
Mokosh II	2	2	0	0	0	2	2
pAgo Long-A	8	8	8	8	8	8	3
qatABCD	2	4	4	2	0	4	0
HEC (GmrSD-like)	19	19	14	14	0	19	19
CRISPR-Cas	42	501	481	104	467	500	394
CBASS	5	24	22	5	24	24	15
AbiD	1	1	0	0	1	1	1
AbiE	16	32	0	0	32	32	0
Aristaios	4	6	0	0	6	3	6
DarTG	11	22	0	0	22	20	9
Pycsar	2	4	0	0	2	2	1
Rugutis	1	1	0	0	1	0	1
SoFic	31	43	0	0	26	43	0
Stk2	1	1	0	0	1	0	0
pAgo Long-B / Sir2	1	2	0	0	2	2	2
PD	46	77	1	1	1	77	1
Kiwa	5	10	0	0	10	10	0
Ogmios	3	3	0	0	3	3	0
dXTPase	3	3	0	0	3	3	0
ietAS	1	2	0	0	0	2	0
Ceres	28	33	0	0	33	0	0
Druantia	6	26	0	0	20	26	21
Menshen	1	2	0	0	0	2	2
gcu233	15	15	0	0	15	0	0
PDC-M	40	122	0	0	0	122	33
PDC-S	52	408	3	3	5	408	20
DP-only unassigned	52	586	12	12	0	0	586
Former R-M annotations	11	12	1	1	0	4	11

Putative nuclease-associated CDSs · Effector CDSs · Rows overlap

Counts: recorded annotations · Track: row union · —: unavailable

PAD: PADLOC; DF: DefenseFinder; DP: DefensePredictor; REB: REBASE